Jean-Baptiste Saulnier *Quantitative Biologist*

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ORCID: 0009-0009-3099-2307 French citizen

RESEARCH EXPERIENCE

2025 – present **Postdoctoral Fellow**
Institute of Plant and Microbial Biology, Academia Sinica, Taipei, Taiwan
Investigating membrane potential regulation in *Bacillus subtilis* biofilms at both single-cell and community scales. Combining mathematical modeling, custom microfluidic chip fabrication, and multiscale microscopy. Supervisor: Dr. Jian-geng Chiou.

2024 – 2025 **CNRS Postdoctoral Fellow**
Laboratoire de Chimie Bactérienne (LCB), Marseille, France
Studied topological defects in collective patterns of *Myxococcus xanthus* mutants to characterize organizational phenotypes. Supervisor: Dr. Tâm Mignot.

2019 – 2023 **PhD in Quantitative Biology**
Laboratoire de Chimie Bactérienne (LCB), Marseille, France
Elucidated the mechanism of spatial pattern transition between rippling and swarming in *Myxococcus xanthus* during predation, combining mathematical modeling, image analysis, and molecular biology. Supervisors: Dr. Tâm Mignot & Dr. Vincent Calvez.

PUBLICATIONS

2024 **The mechanism of spatial pattern transition in motile bacterial collectives**
J.-B. Saulnier, M. Romanos, J. Schrohe, C. Cuzin, V. Calvez, T. Mignot
Preprint — bioRxiv 10.1101/2024.10.28.620572

2023 **A new modeling approach of *Myxococcus xanthus* bacteria using polarity-based reversals**
H. Bloch, V. Calvez, B. Gaudeul, L. Gouarin, A. Lefebvre-Lepot, T. Mignot, M. Romanos, **J.-B. Saulnier**
CEMRACS Proceedings — HAL hal-04102694

EDUCATION

2018 – 2019 **Master's in Theoretical Physics**
Aix-Marseille Université. Developed a reversal-detection algorithm for *M. xanthus* using tracking and image analysis. Supervisor: Dr. Tâm Mignot.

2016 – 2018 **Master's in Cosmos, Fields and Particles** — *Université de Montpellier*

2013 – 2016 **Bachelor's in Fundamental Physics** — *Université de Montpellier*

2011 – 2013 **DUT in Thermal Engineering and Energy** — *Université de Belfort-Montbéliard*

SUPERVISION, CONFERENCES & TRAINING

Supervision

2023 (6 mo.) Undergraduate student supervision (Jonathan Schrohe) — *SgmX polar protein tracking*

Conferences & Training

2026 Quantitative Microbiology Symposium — *Academia Sinica, Taipei* (poster, talk)

2025 Quantitative Microbiology Symposium — *Academia Sinica, Taipei* (poster)

2024 Frontiers in Interacting Particle Systems — *CIRM, Marseille* (talk)

2024 Mathematical Biology: Collective Behavior and Pattern Formation — *CIRM, Marseille*

2022 CENTURI Hackathon: Hacking Living Systems — *CIRM, Marseille*
2022 EMBO Workshop: Adherent Microbial Communities — *Cargèse* (poster)
2022 CEMRACS — *CIRM, Marseille* (talk)
2021 Fall School in Complex Systems — *Nice & Fréjus*

SKILLS

Modeling: PDE, agent-based simulations · Microscopy: phase contrast, epifluorescence · Microfluidics: wafer to chip (photolithography, PDMS) · Programming: Python, SageMath · Languages: English (fluent), French (native)